NumPy Arithmetic Statistical and String Functions

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Start coding or generate with AI.
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Agenda

- Arithmetic Operations Using NumPy
  - Addition
  - Subtraction
  - Multiplication
  - Division
  - Power of
- Statistical Function in Numpy
  - Calculating Median, Mean, Standard Deviation, and Variance in the Array
  - Calculating Percentiles
- String Function in Numpy

1. Arithmetic Operations Using NumPy

1.1 Addition

```
# Perform element-wise addition of two arrays using the 'np.add' method
import numpy as np
a = np.array([30,20,10])
b = np.array([10,20,30])
result = np.add (a,b)
print(result)
```

```
[40 40 40]
```

1.2 Subtraction

```
# Perform element-wise subtraction of two arrays using 'np.subtract' method
a = np.array([[30,40,60], [50,70,90]])
b = np.array([[10,20,30], [40,30,80]])
result = np.subtract (a,b)
print(result)
```

```
[[20 20 30]
 [10 40 10]]
```

1.3 Multiplication

```
# Perform element-wise multiplication of two arrays using the 'np.multiply' method
a = np.array([30,20,10])
b = np.array([10,20,30])
result = np.multiply (a,b)
print(result)
```

```
[300 400 300]
```

1.4 Division

```
# Perform element-wise division of two arrays using the 'np.divide' method
a = np.array([[1,2,1],[2,2,3]])
b = np.array([3,4,5])
result = np.divide (a,b)
print(result)
```

```
[[0.33333333 0.5        0.2       ]
 [0.66666667 0.5        0.6       ]]
```

1.5 Power of

```
# Perform element-wise power operation where each element in 'a' is raised to the corresponding element in 'b' using the 'r
a = [2,2,2,2,2]
b = [2,3,4,5,6]
c = np.power(a,b)
print(c)
```

```
[ 4  8 16 32 64]
```

2. Statistical Function in Numpy

2.1 Calculating Median, Mean, Standard Deviation, and Variance in the Array

```
# Median
array = np.array([[4,3,2],[10,1,0],[5,8,24]])
np.median(array)
print(np.median(array))
# Mean
np.mean(array)
print(np.mean(array))
# Standard Deviation
np.std(array)
print(np.std(array))
# Variance
np.var(array)
print(np.var(array))
```

```
4.0
6.333333333333333
6.944222218666553
48.22222222222222
```

2.2 Calculating Percentiles

- The `np.percentile()` function is used to compute the nth percentile of the array elements.
- The nth percentile value should be in between 0 and 100.

```
# Find the value under the 50th percentile of the NumPy array.
np.percentile(array,50)
```

```
4.0
```

3. String Function in Numpy

```
# Perform element-wise string concatenation for two arrays of string
x = np.array(['Hello','World'])
y = np.array(['Welcome', 'Learners'])
result = np.char.add(x,y)
print(result)
```

```
['HelloWelcome' 'WorldLearners']
```

```
# Replacing the old substring with the new substring
str = "Hello How Are You"
print(str)
z = np.char.replace (str, 'Hello', 'Hi')
print (z)
```

```
Hello How Are You
Hi How Are You
```

```
# Converting all lowercase characters in a string to uppercase and vice-versa
str1 = "hello how are you"
print(str1)
z = np.char.upper (str1)
print(z)
```

```
str2 = "GREETINGS OF THE DAY"
print(str2)
s = np.char.lower (str2)
print(s)
```

```
hello how are you
HELLO HOW ARE YOU
GREETINGS OF THE DAY
greetings of the day
```